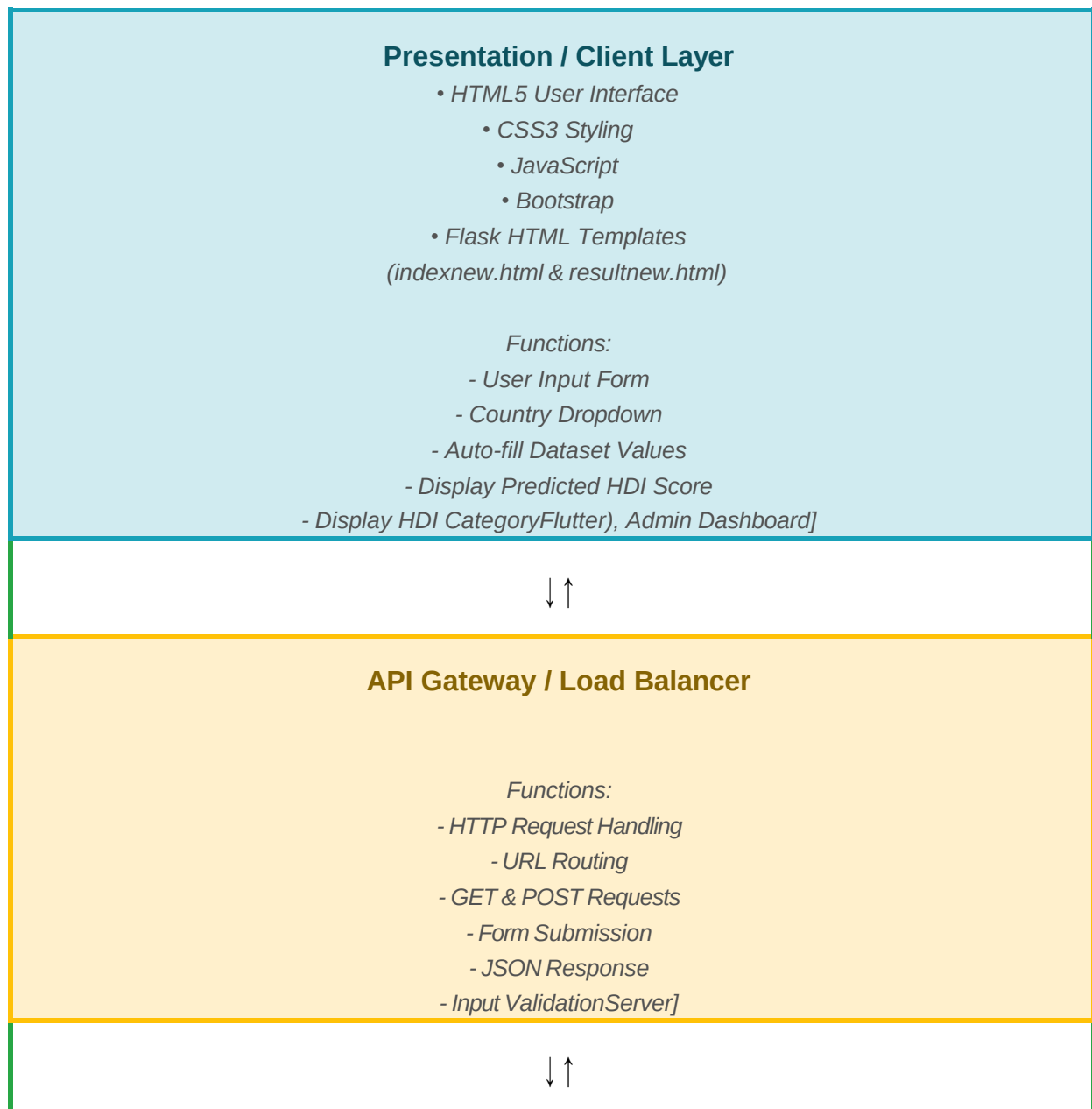


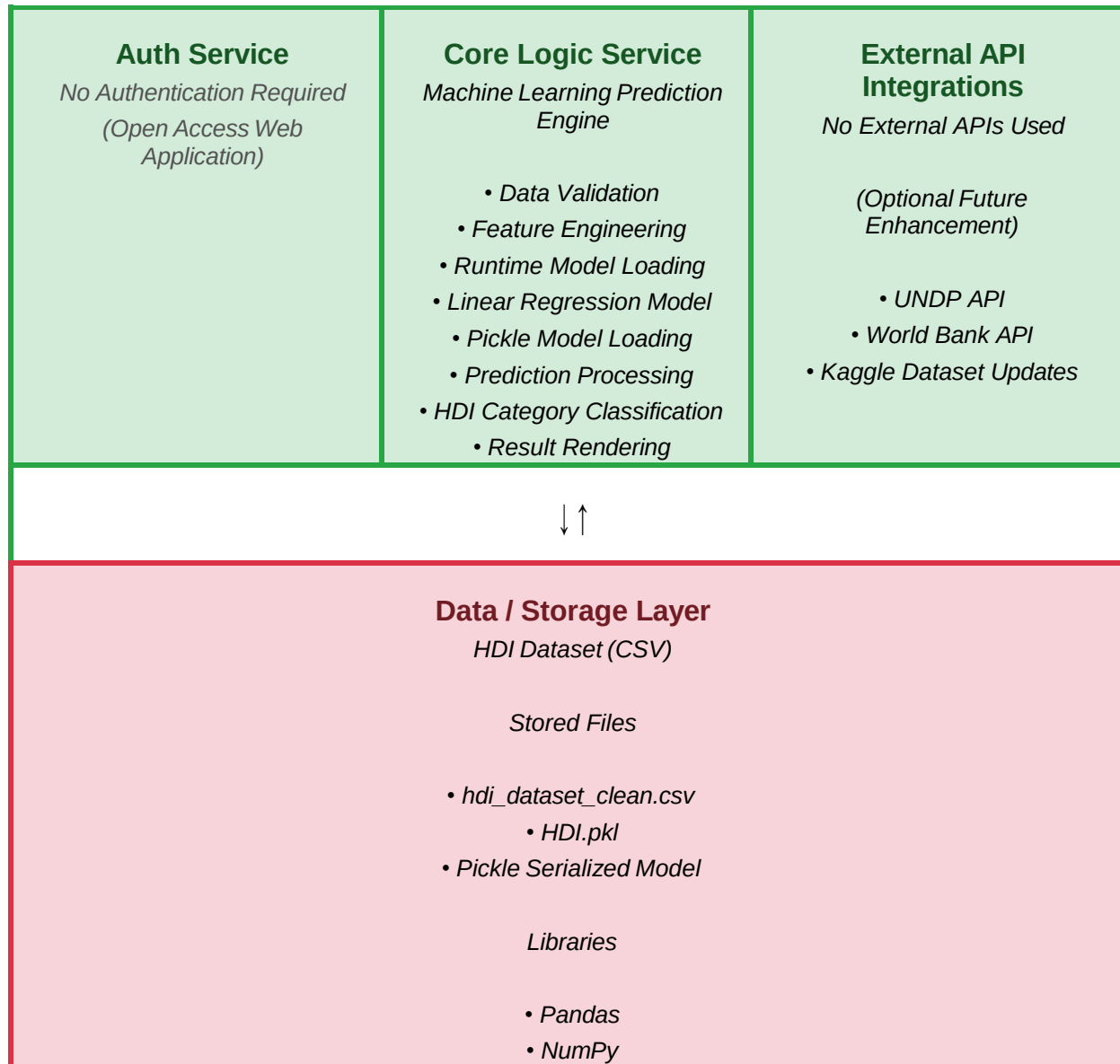
Solution Architecture

Date	
Team ID	
Project Name	A Comprehension of Measure of Well-Being
Maximum Marks	5 Marks

Solution Architecture Diagram:

Fill each box in the architecture diagram with the following text





Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	Provides a user-friendly interface where users select a country or enter socio-economic indicators and view the predicted HDI score and category.	<i>HTML5,CSS3,Java script,Bootstrap</i>
<i>APL Gateway</i>	Handles HTTP GET and POST requests, performs URL routing, validates user inputs, processes requests, and communicates with the prediction engine.	<i>Flask</i>
<i>Core Logic Service</i>	Loads the trained Linear Regression model, preprocesses user inputs, predicts the HDI score, classifies it into Low, Medium, High, or Very High HDI, and sends the results to the frontend	<i>Python,Scikit-learn,Pickle,Pandas</i>
<i>Data/Storage Layer</i>	Stores the cleaned HDI dataset, serialized machine learning model, feature means, and prediction artifacts required during runtime.	<i>CSV Dataset,Pickle</i>